

Two-Period Model: Zoning and Transportation Investment

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Purpose of the Model

Central mechanism

Local zoning restrictions limit the housing and density response to improved accessibility. A transportation agency that takes zoning as given may therefore find transit investment less attractive than it would under coordinated zoning and transportation policy.

- There are two periods in the model.
- In the first period, zoning is chosen locally by municipalities who cares about incumbent interests.
- Then transportation investment is chosen at the metropolitan level taking zoning as given.
- In the second period, Households sort across residence, workplace.
- The economy is closed: welfare changes come from the spatial allocation of residents, jobs, housing prices, wages, congestion, and accessibility.

Basic Environment

- A metropolitan area is partitioned into municipalities $g \in G$.
- Municipality g contains locations S^g :

$$S = \bigcup_{g \in G} S^g, \quad S^g \cap S^{g'} = \emptyset \quad \text{for } g \neq g'.$$

- Each location i has a zoning capacity $h_{0,i}$.
- Higher $h_{0,i}$ means looser zoning and greater housing capacity.
- Transportation investment I_{jk} on link (j, k) reduces commuting costs and improves accessibility.

Policy tension

Zoning is local and politically fragmented, while transportation returns are regional and depend on the entire spatial pattern of density.

Key Agents

Municipal landowners

- Choose local housing capacity $h_{0,i}$
- Represent incumbent interests
- Take other municipalities' zoning as given

Transportation agency

- Observes full zoning vector h_0
- Chooses network investment I
- Maximizes regional household welfare subject to a budget

Households

- Begin at origin o
- Choose residence i
- Choose workplace j
- Face housing prices, wages, moving costs, commuting costs, and congestion

Timing

Time $t = 0$

- 1 Each municipality g chooses $\{h_{0,i}\}_{i \in S^g}$.
- 2 Municipalities play a Nash game in zoning.
- 3 The transportation agency observes h_0 .
- 4 The agency chooses optimal network investment I .

Time $t = 1$

- 1 Households choose residence i , workplace j , moving frictions involved
- 2 Housing markets clear and prices q_i are determined.
- 3 Labor markets clear and wages w_j are determined.
- 4 Local populations N_i are determined.

Municipal Zoning Choice

A representative landowner in municipality g chooses local housing capacity:

$$u_{0,g} = \max_{\{h_{0,i}\}_{i \in S^g}} \sum_{i \in S^g} E\{u_{1,i}(h_{0,i}; h_{0,-g}, I)\} - \lambda \sum_{i \in S^g} h_{0,i}.$$

- $\lambda > 0$ is the per-unit cost of expanding housing capacity.
- Each municipality values incumbent period-1 welfare.
- Each municipality takes other municipalities' zoning choices as given.
- In the decentralized equilibrium, municipalities do not internalize the full regional effect of their zoning choices on housing affordability or transportation returns.

Household Period-1 Problem

A household ω begins at origin o and chooses residence i and workplace j :

$$u_{1,\omega oij} = \max_{c_{ij}, h_{ij}} \frac{B_i}{D_{oij}(I, N)} \left(\frac{c_{ij}}{\alpha} \right)^\alpha \left(\frac{h_{ij}}{1 - \alpha} \right)^{1-\alpha} v_{\omega ij}$$

subject to

$$c_{ij} + h_{ij}q_i \leq w_j - \phi(N_i, h_{0,i}) + h_{0,o}q_o.$$

- B_i : residential amenity at destination i .
- q_i : housing price at residence location i .
- w_j : wage at workplace location j .
- $h_{0,o}q_o$: rental income from origin housing owned by the incumbent.
- $\phi(N_i, h_{0,i})$: local congestion/crowding cost.

Congestion, Moving, and Commuting Frictions

The spatial friction for a household moving from o to residence i and workplace j is

$$D_{oij}(I, N) = \varepsilon_{oij} \exp(\rho m_{oi} + \kappa \tau_{ij}(I, N)).$$

- ε_{oij} : deterministic bilateral component.
- m_{oi} : moving cost from origin o to residence i .
- $\tau_{ij}(I, N)$: commuting cost from residence i to workplace j .
- Commuting costs decrease with investment and increase with congestion.
 $\partial \tau / \partial N \geq 0$ (congestion) and $\partial \tau / \partial I \leq 0$ (investment reduces cost).

Preference Heterogeneity

Households draw idiosyncratic preferences over residence-workplace pairs:

$$\Pr(v_{\omega ij} \leq z) = \exp(-T_i E_j z^{-\epsilon}), \quad z > 0, \quad \epsilon > 1.$$

- T_i captures residence attractiveness at location i .
- E_j captures workplace attractiveness at location j .
- ϵ governs dispersion in idiosyncratic preferences.
- The Fréchet structure gives tractable sorting across residence-workplace pairs.

Transportation Agency Problem

In period 0, after observing the zoning vector h_0 , the agency chooses network investment:

$$\max_{\{I_{jk}\}} \sum_o N_o^0 \bar{U}_o(I; h_0)$$

subject to

$$\sum_{j,k} \delta_{jk} I_{jk} \leq K, \quad \underline{I}_{jk} \leq I_{jk} \leq \bar{I}_{jk}.$$

Transportation investment enters commuting costs:

$$\tau_{jk}(I, N) = \delta_{jk}^{\tau} \frac{N_{jk}^{\rho}}{I_{jk}^{\gamma}}, \quad \frac{\partial \tau}{\partial N} \geq 0, \quad \frac{\partial \tau}{\partial I} \leq 0.$$

The agency is welfare-oriented, but it treats local zoning capacity as fixed.

Decentralized Equilibrium

A decentralized equilibrium is a profile

$$(\{h_{0,i}^*\}, l^*, \{\pi_{oij}^*\}, \{q_i^*, w_j^*\})$$

such that:

- 1 Households choose residence-workplace pairs optimally.
- 2 Housing and labor markets clear at (q_i^*, w_j^*) .
- 3 Each municipality chooses zoning capacity optimally, taking other municipalities and transportation policy as given.
- 4 The transportation agency chooses l^* to maximize regional welfare conditional on h_0^* .

Source of distortion

The municipality's zoning choice reflects incumbent welfare within its jurisdiction, not the full metropolitan benefits from cheaper housing, better sorting, or stronger transit returns.

What Local Zoning Omits

The decentralized municipal choice omits two positive regional externalities:

1. Cross-municipality welfare externality

Looser zoning at i lowers local housing prices and makes location i more accessible to households from other municipalities. The local municipality does not fully value these gains.

2. Transit-zoning externality

Looser zoning at i raises density near accessible locations. Higher density can raise transit ridership and increase the marginal return to transit investment. A local municipality does not internalize how its zoning affects the regional network choice.

Main implication

Decentralized zoning is over-restrictive relative to the coordinated metropolitan optimum:

$$h_{0,i}^{\text{SP}} > h_{0,i}^{\text{DE}}.$$

Social Planner Benchmark

The planner jointly chooses zoning and transportation investment:

$$\max_{\{h_{0,i}\}, \{I_{jk}\}} \sum_o N_o^0 \bar{U}_o(h_0, I)$$

subject to

$$\sum_{j,k} \delta_{jk} I_{jk} \leq K, \quad h_{0,i} \geq 0.$$

- The planner internalizes the housing-price effect of zoning across all households.
- The planner internalizes how zoning changes optimal transportation investment.
- The planner can coordinate density and transit investment in the same locations.

Benchmark comparison

The welfare gap between the decentralized equilibrium and the planner allocation measures the cost of fragmented land-use control.